Course: IT160IU - Data Mining

Data Mining Project

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Introduction

Objective

Build a data mining framework with classification/prediction and clustering/sequence analysis

Dataset

Heart Disease (Kaggle)

Data Preprocessing

Raw Data

- 10,000 instances
- 21 attributes

Tasks

- Missing values
- Duplicates
- Outliers

Target

- Heart Disease Status(Yes / No)

Transformation

- Feature–target split
- Encoding categorical variables

Data Cleaning

Handling missing value

	Missing_Count	Missing_%
Alcohol Consumption	2586	25.86
Diabetes	30	0.30
Sugar Consumption	30	0.30
Cholesterol Level	30	0.30
Age	29	0.29
Triglyceride Level	26	0.26
CRP Level	26	0.26
High LDL Cholesterol	26	0.26
High Blood Pressure	26	0.26
Low HDL Cholesterol	25	0.25
Sleep Hours	25	0.25
Exercise Habits	25	0.25
Smoking	25	0.25
Fasting Blood Sugar	22	0.22
BMI	22	0.22
Stress Level	22	0.22
Family Heart Disease	21	0.21
Homocysteine Level	20	0.20
Blood Pressure	19	0.19
Gender	19	0.19
Heart Disease Status	0	0.00

Total missing values: 3054

Data Cleaning

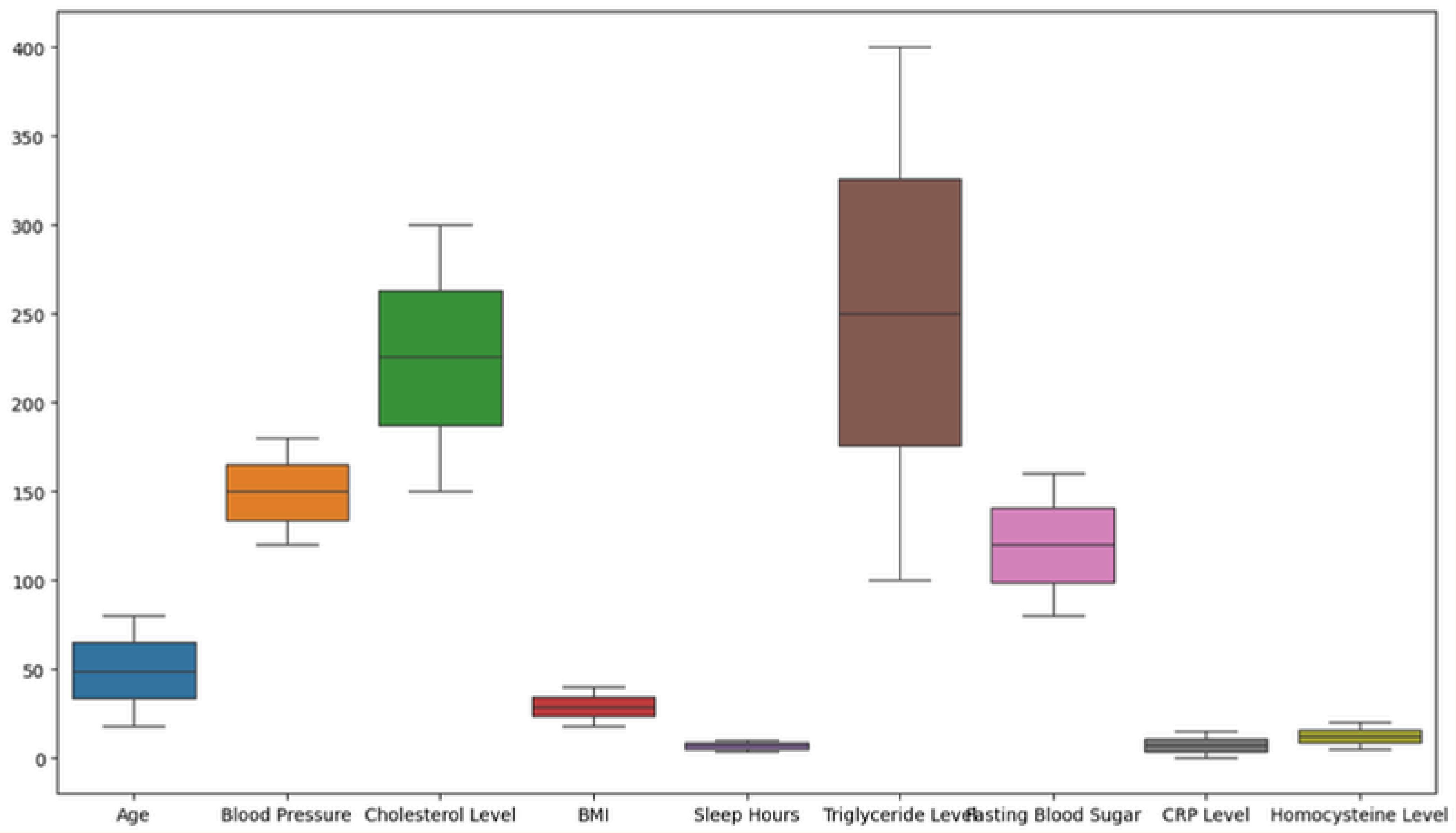
Duplicate Records

```
df_clean = df.drop_duplicates().copy()
print("Shape after dropping duplicates:", df_clean.shape)

for col in df_clean.select_dtypes(include="object").columns:
    print(f"{col}: {df_clean[col].unique()[:15]}")

cat_cols = df_clean.select_dtypes(include="object").columns

for col in cat_cols:
    df_clean[col] = df_clean[col].astype(str).str.strip().str.title()
```

```
Shape after dropping duplicates: (10000, 21)
Gender: ['Male' 'Female' nan]
Exercise Habits: ['High' 'Low' 'Medium' nan]
Smoking: ['Yes' 'No' nan]
Family Heart Disease: ['Yes' 'No' nan]
Diabetes: ['No' 'Yes' nan]
High Blood Pressure: ['Yes' 'No' nan]
Low HDL Cholesterol: ['Yes' 'No' nan]
High LDL Cholesterol: ['No' 'Yes' nan]
Alcohol Consumption: ['High' 'Medium' 'Low' nan]
Stress Level: ['Medium' 'High' 'Low' nan]
Sugar Consumption: ['Medium' 'Low' 'High' nan]
Heart Disease Status: ['No' 'Yes']
```

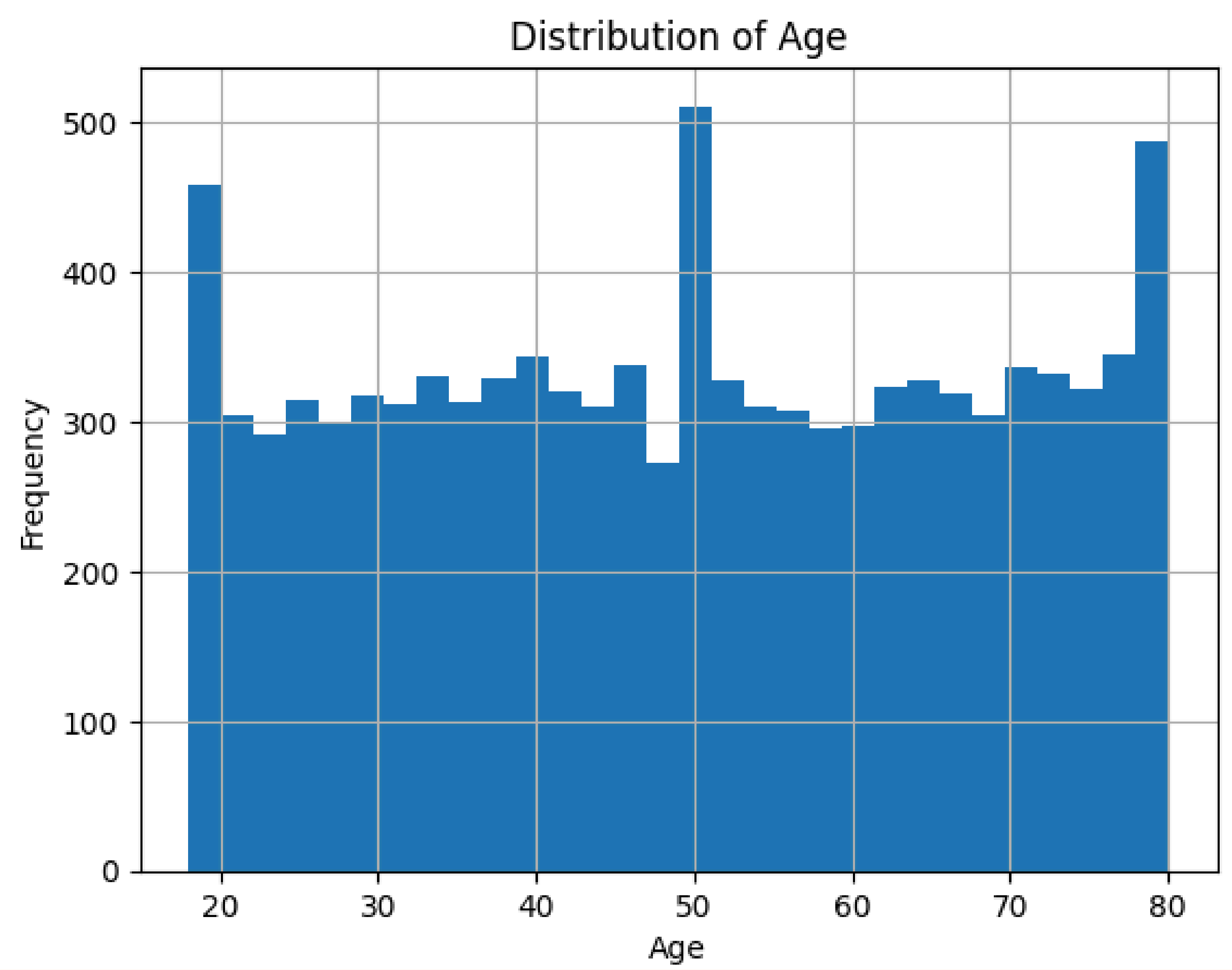
Choose Label & Handle Outlier

Outlier Inspect



Data Analyst

Exploratory Data Analysis (EDA)



Data Analyst

Cross-tab analysis

- Gender
- Exercise Habits
- Smoking Behavior
- Family Heart Disease History
- Stress Level

=== Gender vs Heart Disease Status ===

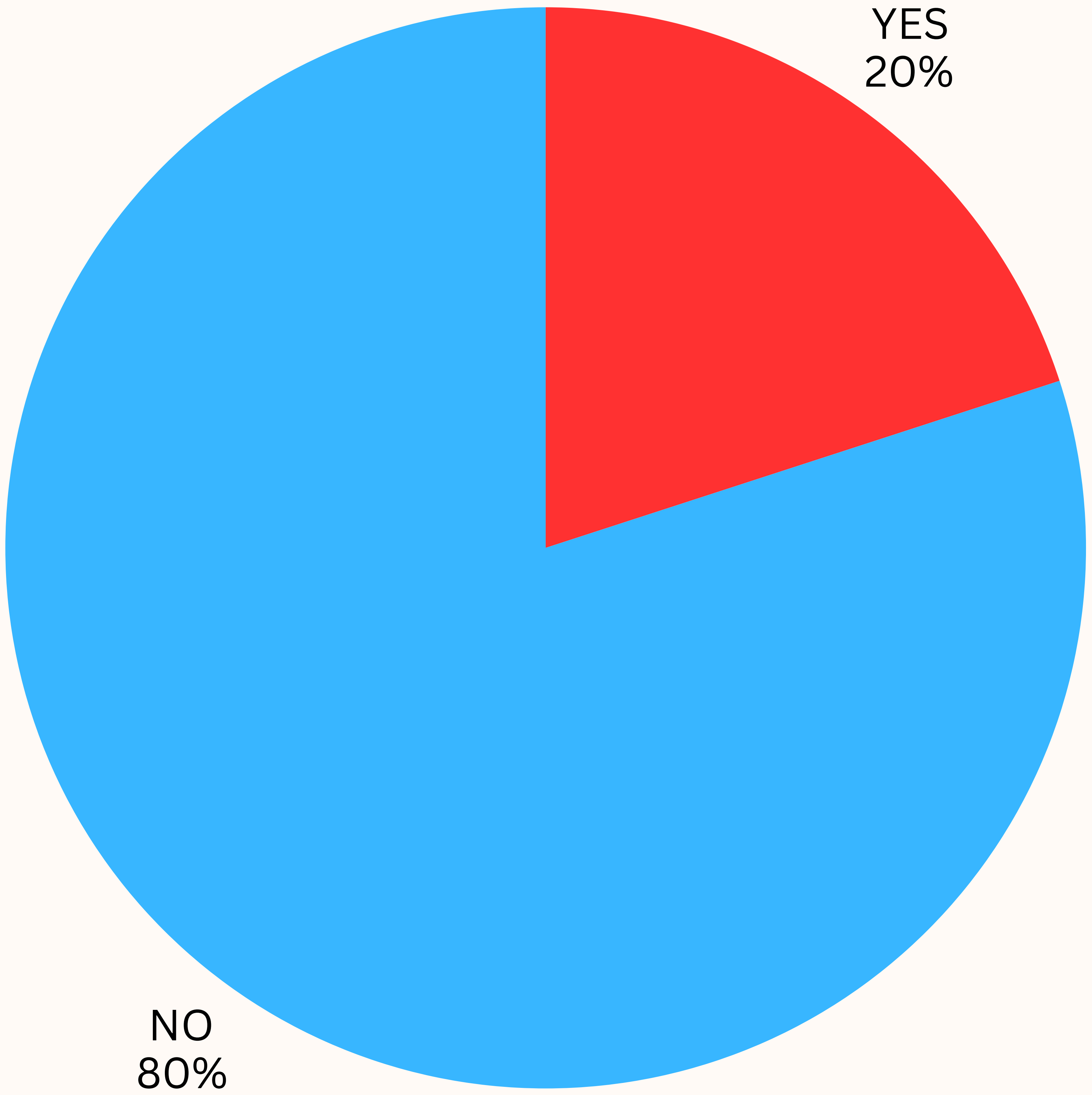
Heart Disease Status	No	Yes
Gender		
Female	79.31	20.69
Male	80.63	19.37
Nan	94.74	5.26

=== Exercise Habits vs Heart Disease Status ===

Heart Disease Status	No	Yes
Exercise Habits		
High	79.98	20.02
Low	80.28	19.72
Medium	79.62	20.38
Nan	96.00	4.00

Data Analyst

Class Imbalance Assessment



Classification Model

For “Step 2” process

- J48 Decision Tree (C4.5)
- Random Forest
- Naive Bayes
- Logistic Regression

For “Step 3” process

- Cost-Sensitive Learning with Random Forest
- AdaBoost (Adaptive Boosting)
- SMO (Sequential Minimal Optimization)

Model Comparison

“Step 2" Process:

- Baseline Performance
- Standard, interpretable classifiers (e.g., J48, Naive Bayes).
- Lower computational cost; easier to explain.
- Standard classification approach.

“Step 3" Process:

- Performance Optimization
- Advanced, ensemble, & optimization methods (e.g., AdaBoost, SMO).
- Higher complexity; focuses on maximizing accuracy/handling costs.
- Specifically addresses imbalances (Cost-Sensitive Learning).

Model Comparison

Model	Accuracy	Recall (Positive)	Precision (Positive)	F1-Score	AUC
Baseline Models (Step 2)					
J48	~72.7%	0.134	0.211	0.164	0.509
Naive Bayes	~70%	~0.35	~0.20	~0.26	~0.50
Random Forest (baseline)	~80%	**0.00**	—	—	0.498
Logistic Regression	~72–73%	~0.30	~0.20	~0.24	~0.50
Improved Models (Step 4)					
SMO (Resample inside CV)	~50.9%	**0.479**	0.198	0.280	0.498
AdaBoostM1 (Resample inside CV)	~43.7%	**0.599**	0.199	0.298	0.496
Cost-Sensitive Random Forest	~30.6%	**0.827**	0.201	**0.323**	0.501

Model Evaluation

All models were evaluated on the same preprocessed dataset (heart_clean.arff).
The evaluation metrics collected include:

- Accuracy
- Precision, Recall, F1-score for the minority (positive) class
- AUC (Area Under ROC Curve)

Runtime required to train and evaluate each model (in milliseconds)

Model	Accuracy	Recall (Positive)	Precision (Positive)	F1-Score	AUC	Runtime (ms)
J48 (baseline)	~72.7%	0.134	0.211	0.164	0.509	2567
Naive Bayes (baseline)	~70%	~0.35	~0.20	~0.26	~0.50	184
Random Forest (baseline)	~80%	0.000	—	—	0.498	52041
Logistic Regression (baseline)	~73%	~0.30	~0.20	~0.24	~0.50	555
SMO (Resample Inside CV)	~50.9%	0.479	0.198	0.280	0.498	11204
AdaBoost M1 (Resample Inside CV)	~43.7%	0.599	0.199	0.298	0.496	6500
Cost-Sensitive Random Forest	~30.6%	**0.827**	0.201	**0.323**	0.501	102309

Conclusions

- **Key Challenge**: Highly imbalanced data (20% positive cases).
- **Initial Finding**: Baseline models had high accuracy but failed to detect disease (near-zero Recall).
- **Solution**: Applied Cost-Sensitive Learning and AdaBoost to prioritize minority class detection.
- **Final Result**: Cost-Sensitive Random Forest proved best, boosting Recall to 0.83.
- **Future Work**: Feature engineering, PCA, and Deep Learning exploration.

References

https://www.kaggle.com/datasets/oktayrdeki/heart-disease
https://www.cs.waikato.ac.nz/ml/weka/
https://weka.sourceforge.io/doc.dev/
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THANK YOU